

Algebraic topology (MA4104) presentation: Classifying all connected double and triple covers of figure eight upto isomorphism

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Let X be a topological space. We begin by defining the notion of a covering space.

A topological space $\tilde{X}$ is said to be a covering space of X if there exists a continuous surjective map $p : \tilde{X} \rightarrow X$ such that for every $x \in X$, there exists a neighbourhood $U_x \ni x$ such that

1. $p^{-1}(U_x) = \bigsqcup_{\alpha \in I} V_{\alpha}^x$, where V_{α}^x 's are open sets in $\tilde{X}$, and,
2. $p|_{V_{\alpha}^x} : V_{\alpha}^x \rightarrow U_x$ is a homeomorphism.

Such neighbourhoods U_x are said to be evenly covered neighbourhoods of X and the map p is said to be a covering map.

We want to classify all double and triple covers of the "figure eight", which is as drawn below.



Figure 1: Figure eight.

Let us call this space as $(X, x_0) (\subset \mathbb{R}^2)$ where $x_0 = (0, 0)$ is the based point of X as drawn in the picture. Clearly, this space X is a wedge of two circles $(S^1 + (1, 0))$ and $(S^1 + (-1, 0))$, which are generated by two loops, let say a and b based at x_0 .

Note that if we choose any point $x \in X$ other than the base point itself, then any evenly covered neighbourhood of x will be homeomorphic to an open interval in $\mathbb{R}$. So, if we choose such an evenly covered neighbourhood U_x , then $p^{-1}(U_x)$ is just a disjoint union of open intervals in $\mathbb{R}$. Now, considering the base point x_0 , it is easy to see (by observing figure 1) that any evenly covered neighbourhood U_{x_0} of x_0 will consist of the vertex x_0 along with four edges, wherein two edges each of a and b are oriented towards the vertex, and the other two edges each of a and b are oriented away from the vertex.

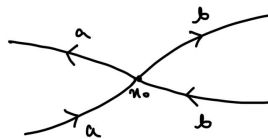


Figure 2: Evenly covered neighbourhood U_{x_0} .

From now on, we call any vertex having a neighbourhood like that of x_0 (see fig. 2) to be a **special vertex**. To satisfy the above mentioned property 2 of a covering space, we observe that all points in the fibre of x_0 should be special vertices. Also, we will use the notion of isomorphism of covers, which means that if $p_1 : \tilde{X}_1 \rightarrow X$ and $p_2 : \tilde{X}_2 \rightarrow X$ are two covering maps of X , then, $\tilde{X}_1$ and $\tilde{X}_2$ are said to be isomorphic as covers if there exist a homeomorphism $\phi : \tilde{X}_1 \rightarrow \tilde{X}_2$ such that the following diagram commutes,

$$\begin{array}{ccc} \tilde{X}_1 & \xrightarrow{\phi} & \tilde{X}_2 \\ p_1 \searrow & & \swarrow p_2 \\ & X & \end{array}$$

$$\text{i.e. } p_2 \circ \phi = p_1.$$

With these information in hand, let us start classifying all the possible double and triple covers of figure eight up to isomorphism.

1 Double covers of figure eight

By a double cover, we mean that $|p^{-1}(x_0)| = 2$, that is, our covering space should consist of two special vertices. Let α, β be these two special vertices.

Now, at each vertex α and β , the loop b lifts either to a loop or to a path. But, since, the covering space consists of just two vertices, it is easy to see that if the loop b lifts to a loop at one vertex, then at the other vertex also, it has to lift to a loop. It is because if the loop lifts to a loop at let say, α , then, at the vertex β , if it lifts to a path, the end point of this path should be the vertex α . But this would then imply that there are two b -edges oriented towards α , which gives a contradiction to the fact that α is a special vertex. Again using the same logic, one can see that if b lifts to a path at one special vertex, then, at the other special vertex also it has to lift to a path. So, we just have the following two choices of covering spaces showing the lifts of b :



Figure 3: Only possible covering spaces showing just the lift of b .

Now, we need to lift a in a manner so as the covering space remains connected. In the covering space (1), if the loop a lifts to loops at both the special vertices α and β , then we see that the covering space becomes disconnected. Henceforth, at least at one vertex, moreover by the logic we mentioned above in the case of the lifts of b , at both vertices, a lifts to a path. In the covering space (2), we can lift a in any manner since the vertices α and β are already connected. So by the same argument, the only possible choices for the lifts of a will be same as that of the lifts of b as shown in fig.3. This leads us to conclude that there are only three possible choices for a two-sheeted connected covering space of figure eight, and they are as shown in the next page. Thus, **upto isomorphism, there exist only three connected double covers of figure eight.**

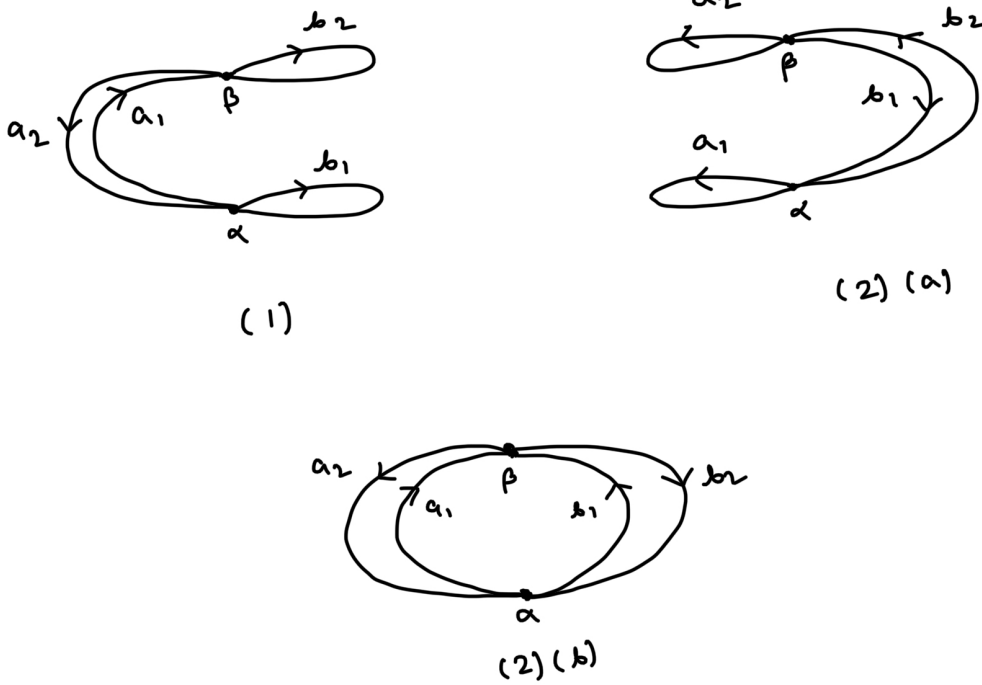


Figure 4: Connected double covers of figure eight.

2 Triple covers of figure eight

In this case, we have $|p^{-1}(x_0)| = 3$, that is, our covering space will consist of three special vertices, so let us name them as α, β, γ . The strategy that we will acquire to figure out all the triple covers of figure eight up to isomorphism is very similar to the case of double cover. We will first see the no. of ways we can lift the loop b to these three vertices without any constraints, and then we will try to lift the loop a in a manner so as the covering space remains connected. (We could have also done it by seeing the no. of ways we can lift the loop a without any constraints, and then try to lift b by keeping the space connected. But both of them will give us the same set of covers, so without loss of generality, we can start with any cover.)

By uniqueness of path lifting, there are three unique lifts of b starting at α, β, γ respectively. Now, these lifts of b can be such that either (see next page fig.5 for reference)

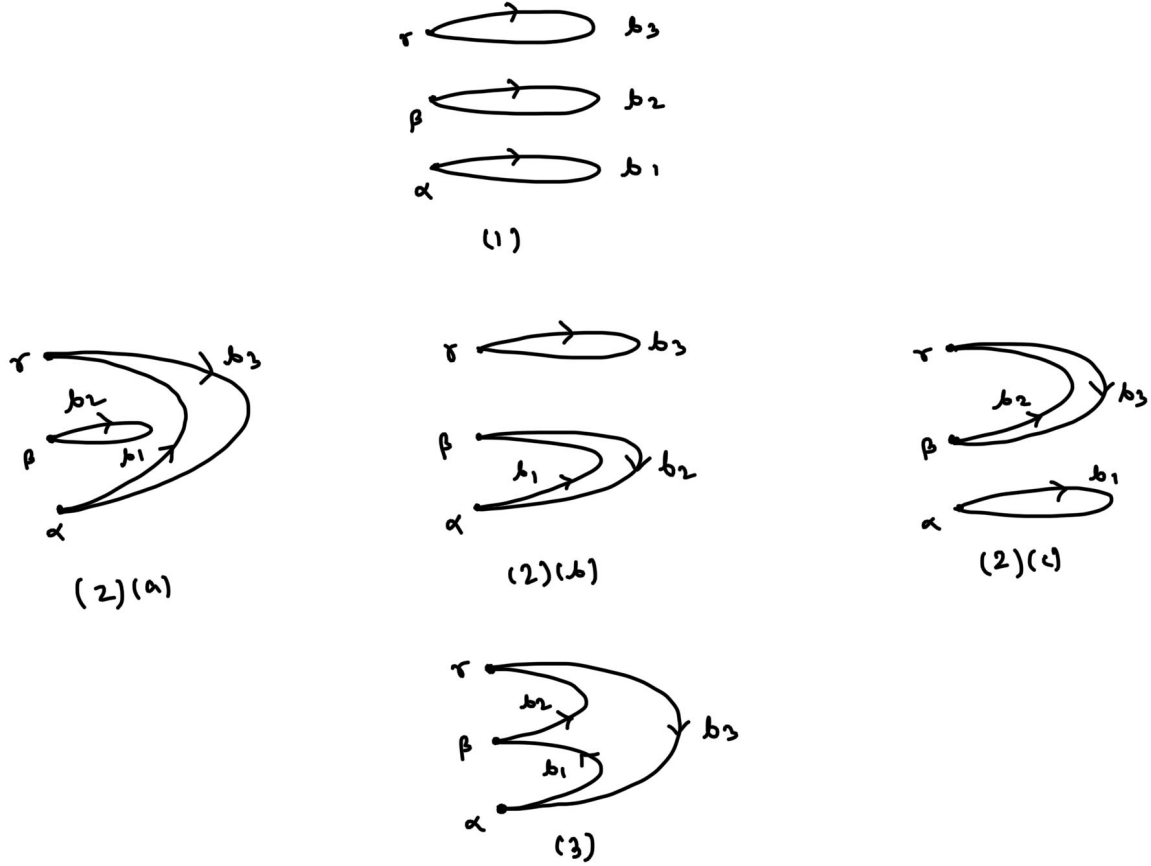
1. none of the three special vertices are connected by a lift of b , i.e. all the three lifts are loops, or,
2. exactly two of the special vertices are connected with each other, i.e. exactly two of the lifts of b are paths and the third one is a loop, or,
3. all the three special vertices are connected with each other, i.e. all the three lifts of b are paths.

Note that in fig.5, (2)(a), (2)(b), (2)(c) are all isomorphic. So, to get all the triple covers up to isomorphism, we only work with (1), (2)(a), which we rename as (2), and (3).

Let us first start with (1). Since, in (1), all the lifts of b are loops, and we want a connected covering space, it follows that all the lifts of a should be paths, because if not, at least one of the special vertices will have a loop as a lift of a , and henceforth, will become disconnected from the rest of the covering space. So, we obtain our first triple cover of figure eight, where all the lifts of b are loops and all the lifts of a are paths. (refer fig.6)

Now, for (2), we divide the problem into two categories: The loop a lifts to

- i only paths (i.e. no loops) at all the three vertices.

Figure 5: Only possible triple covering spaces showing just the lift of b .

ii a loop in at least one of the vertices.

2)i) If the loop a lifts to paths at all the three vertices, then, this gives us our second connected triple cover of the figure eight. (refer fig.6)

2)ii) Note that the loop a cannot lift to a loop at the vertex β , otherwise again the space will be disconnected, but, as of now it can lift to a loop at the vertex α and γ . Suppose it lifts to a loop at α , then, it cannot lift to a loop at γ , because the path, which is a lift of a , at the vertex β should have its end point at either α or γ , but to preserve the special vertex structure of α , the only choice for the end point of this path is γ . Again, to preserve the special vertex structure of γ and β , the only choice for a lift of a at the vertex γ is a path ending at the vertex β . This gives us our third triple cover of the figure eight (refer fig.6), and also proves that a cannot be simultaneously lifted to exactly two loops. Now, if we lift a to a loop at the vertex γ , then, we can follow the same procedure to get a covering space which will be isomorphic to the one we got previously by lifting a to a loop at α .

Let us now move on to (3). We can lift a in whichever manner (obviously preserving the special vertex structure of the vertices) we want, as the space will remain connected because the three lifts of b already connect all the three vertices. We again divide this case into three sub-cases as follows:

i all the lifts of a are loops,

ii exactly one of the lifts of a is a loop, and,

iii all the lifts of a are paths.

The above three covers all the cases because we've already seen that because of the special vertex structure, a loop cannot lift to two loops simultaneously, hence, it has to either lift to one loop, or three loops, or all paths. For

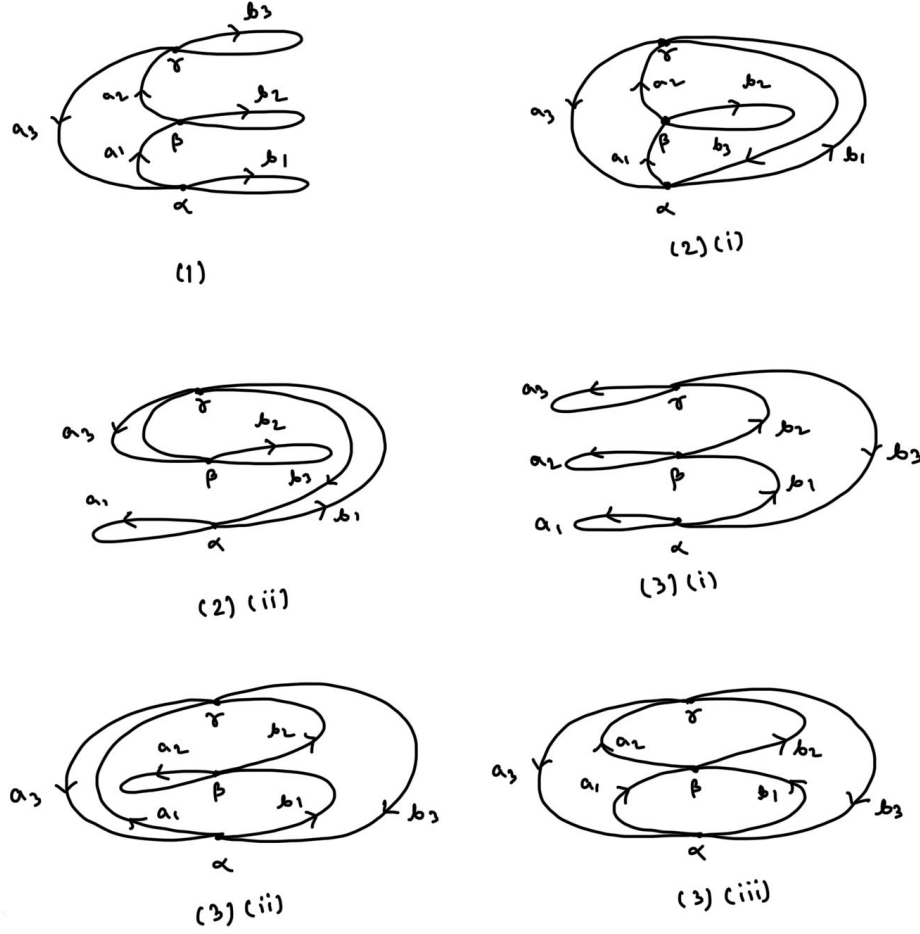


Figure 6: All connected triple covers of figure eight up to isomorphism.

all the cases 3)i), 3)ii), 3)iii), we get three distinct connected triple covers of figure eight (refer fig.6).

Clearly, all of the above spaces in fig.6 are covering spaces for figure eight, because every point $x \in X$ has an evenly covered neighbourhood wrt each of these six covering spaces. Also, by construction, all of these six covering spaces are non-isomorphic, because if we consider any two pair of these covering spaces, then there exist at least one lift of either a or b such that it is a loop in one of the spaces in the pair, and it is a path in the other, which violates the definition of isomorphism of covers as stated earlier. Henceforth, **up to isomorphism, there exists six connected triple covers of figure eight.**
